Candidate No.__________________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Computer Applications (MCA)
MCA (Lateral) Semester-II /MCA Semester-IV
University Examination May 2018
CA841 – Software Quality Assurance (SQA)

Date: 28.05.18, Monday **Time:** 10:00 a.m. to 01:00 p.m. **Maximum Marks:** 70

Instructions:

1. Answer to the two sections must be written in the separate answer books.
2. Answer to different questions should start on new page.
3. Figure to the right indicate full marks.
4. Make suitable assumptions wherever necessary & mention them clearly.

SECTION – I

Q-1 Do as directed.

(a) Answer the following multi choice questions. **07**

1. In order to generate effective tests cases, which are the sources of information
   a) Requirements b) Source code c) Inputs and outputs d) All of these
2. ________ testing is concerned with what the system does its features or functions.
   a) Functional b) Non Functional c) Both a & b d) None of these
3. Let the computer find the error philosophy is used in________ approach of debugging.
   a) Backtracking b) Brute Force c) Cause Elimination d) All of these
4. A program unit can be viewed as having a well-defined________ point.
   a) Entry b) Exit c) Both a & b d) None of these
5. ________ coverage refers to executing individual program statements and observing the outcome.
   a) Statement b) Branch c) Predicate d) None of these
6. In ________ integration testing, all or most of the units are combined together and tested at one go.
   a) Top Down b) Big Bang c) Bottom Up d) Sandwich
7. The main characteristics of agile software development processes are________ .
   a) Incremental development b) Frequent regression testing c) Writing test code one test case at a time before the final code. d) All of these

(b) State True/False. **04**

1. Errors found in unit testing can be fixed easily.
2. In any organization, Decisions and Plans are developed based on the intuition and experience.
3. Testing plays an important role in achieving and assessing the quality of a software product.
4. A test case is a simple pair of input and expected outcome.

Q-2 Answer the following questions.

(a) Explain Static Unit Testing in detail. **06**

(b) Draw the Control Flow Graph for a module to return sum of all even numbers from 1 to n. **06**

OR

Q-2 Answer the following questions.

(a) What is Mutation Testing? How can we calculate mutation score? Show it with example. **06**

(b) Draw the Data Flow Graph for a module to return sum of all odd numbers from 1 to n. **06**

Q-3 Answer the following questions. (Any Three)

(a) List the types of Integration Testing. Explain Big Bang Integration Testing with example. **04**

(b) Differentiate between Validation and Verification. **04**

(c) Explain Equivalence Class Partition by taking appropriate example. **04**

(d) Explain how test cases can be formed using Decision Tables. **04**

SECTION – II

Q-4 Do as directed.

(a) Answer the following multi choice questions. **07**

1. Behavioral testing is ________ testing.
   a) White Box b) Black Box c) Grey box d) None of these
2. ________ system level test provides an evidence that the system can be installed, configured, and brought to an operational state.
   a) Documentation b) Basic c) Load d) Regulatory
3. ________ testing is the first step in the Software Development Life Cycle, where the application is tested as a whole.
   a) Unit b) System c) Integration d) Alpha Testing
4. Boundary value analysis is a________ method.
   a) White Box b) Black Box c) Both a & b d) None of these
5. ________ is the ability to describe and follow the life of a requirement, in both forward and backward direction.
   a) Requirements Traceability b) Coverage Metrics c) Both a & b d) None of these
6. The ________ section of System Test Plan summarizes the high level description of the functionalities of the system.
   a) Introduction b) Assumption c) Feature Description d) Test Approach
7. ________ of test cases means ordering the execution of test cases according to certain test objectives.
   a) Sequencing b) Prioritization c) Sorting d) None of these

(b) State True/False. **04**

1. Pareto principle can be stated as concentrate on the vital few and not the trivial many.
2. The most common error that occurs during integration testing is known as interface error.
3. SQA stands for Software Quality Assessment.
4. In test-driven development, programmers design and implement test cases before the production code is written.

Q-5 Answer the following questions.

(a) Show the estimation of number of test cases by computing the Function Points. **06**

(b) What is severity and priority of a defect? List the metrics used to measure and monitor the defects. Explain the different states of a defect. **06**

OR

Q-5 Answer the following questions.

(a) Explain Alpha and Beta Testing. **06**

(b) Explain the structure of System Test Plan. **06**

Q-6 Answer the following questions. (Any Three)

(a) List different views of Software Quality and explain *any three* in detail. **04**

(b) Explain the McCall's Quality Factors. **04**

(c) Explain *any eight* elements of SQA. **04**

(d) Explain Six Sigma for software engineering. **04**